

Analysis Report: Phi-4 Mini Instruct

Quantized Models

The **Phi-4 Mini Instruct** is a 3.8B parameter model optimized for efficient CPU deployment through GGUF quantization. This compact model supports a 128K token context length and delivers strong reasoning capabilities for our badge generation use case while maintaining modest hardware requirements.

Model Specifications

Specification	Details
Parameters	3.8 billion
Architecture	Dense decoder-only Transformer
Context Length	128K tokens (standard)
Format	GGUF with multiple quantization levels
Original Size (FP16)	~7.6 GB

Quantization Levels & Memory Requirements

Quantization	File Size	RAM Required (Base)*	RAM with 4K Context	RAM with 16K Context	Quality Loss (Perplexity)	Recommended Use
Q2_K	~1.51 GB	~2.0 GB	~2.5 GB	~3.5 GB	Extreme (+0.87 ppl)	Not recommended
Q3_K_M	~2.0 GB	~2.5 GB	~3.0 GB	~4.0 GB	Very high (+0.24 ppl)	Minimal RAM scenarios
Q4_K_M	~2.49 GB	~3.0 GB	~3.5 GB	~4.5 GB	Balanced (+0.05 ppl)	Recommended default

IQ4_XS	~2.22 GB	~2.8 GB	~3.3 GB	~4.3 GB	Good (+0.08 ppl)	Efficient compromise
Q5_K_M	~3.0 GB	~3.5 GB	~4.0 GB	~5.0 GB	Very low (+0.01 ppl)	Balanced quality/size Recommended
Q6_K_L	~3.30 GB	~4.0 GB	~4.5 GB	~5.5 GB	Near-perfect (+0.004 ppl)	Quality priority
Q8_0	~4.08 GB	~4.5 GB	~5.0 GB	~6.0 GB	Minimal (<0.001 ppl)	Maximum quality
F16	~7.6 GB	~8.0 GB	~8.5 GB	~9.5 GB	None	Original quality

*Base RAM includes model weights only; additional RAM needed for KV cache scales with context length

CPU Performance Characteristics

The **CPU usage** for GGUF quantized models depends on several factors including quantization level, context length, and processor architecture.

Aspect	Q4_K_M	Q5_K_M	Q8_0
Inference Speed	Fastest	Moderate	Slowest
Tokens/Second (CPU)	10-15 t/s	8-12 t/s	5-8 t/s
CPU Utilization	60-80%	70-85%	80-95%
Memory Bandwidth	Lower	Moderate	Higher
Recommended CPU	4+ cores	6+ cores	8+ cores

Quantization Method Comparison

GGUF (GPT-Generated Unified Format) is optimized specifically for CPU deployment.

Method	Best For	CPU Performance	Precision	Overhead
GGUF Q4_K_M	CPU inference	Excellent	Good	Low
GGUF Q5_K_M	Balanced	Very Good	Better	Moderate
GGUF Q8_0	Quality priority	Good	Excellent	Higher
GPTQ	GPU only	Poor	Good	N/A
AWQ	GPU only	Poor	Better	N/A

Recommended Configurations by Use Case

Use Case	Quantization	RAM	Context	CPU	Accuracy	Speed
General chatbot	Q4_K_M	8 GB	8K	4-6 cores	99%+	Fast (10-15 t/s)
Document analysis	Q5_K_M	12 GB	32K	6-8 cores	98-99%	Good (6-10 t/s)
Large documents	Q5_K_M	16 GB	64K	8+ cores	97-98%	Moderate (3-5 t/s)
Quality-critical long RAG	Q8_0	24 GB	128K	12+ cores	98-99%	Slower (1-2 t/s)

Quick Decision Matrix

Choose Q4_K_M if: RECOMMENDED

- Context needs are $\leq 16K$ tokens
- RAM limited (8GB available)
- Speed is priority over absolute quality
- Budget/resource constrained deployment

Choose Q5_K_M if: RECOMMENDED

- Context needs are 16-64K tokens
- 12-16GB RAM available
- Best balance of quality, speed, and resource usage needed
- Production deployment requiring reliability

Choose Q8_0 if:

- Context needs are 64K+ tokens
- Quality is critical (research, professional writing)
- 24GB+ RAM available
- Willing to sacrifice speed for accuracy

Cost Estimation (AWS services):

Configuration	50 Users (Avg)	100 Users (Avg)
4 vCPU / 8GB	\$ 1,200/month	\$ 2,100/month
8 vCPU / 16GB	\$ 1,400/month	\$ 2,700/month
12 vCPU / 24GB	\$ 1,600/month	\$ 3,000/month

Note : We have deployed the fine-tuned model with Q4_K_M quantization, 8 vCPU cores-16GB RAM, 6144 context length.

Multilingual Capabilities

Phi-4 Mini Instruct demonstrates strong multilingual capabilities essential for global credentialing systems. The model supports badge generation, prompt engineering, and reasoning tasks across multiple languages with ISO 639-1 language code support integrated into the MIT-DCC badge pipeline.

Supported Languages

23 Languages supported by SLM

Arabic, Chinese, Czech, Danish, Dutch, English, Finnish, French, German, Hebrew, Hungarian, Italian, Japanese, Korean, Norwegian, Polish, Portuguese, Russian, Spanish, Swedish, Thai, Turkish, Ukrainian

Key Terms Explained for Beginners

Core Concepts

Quantization

The process of compressing a language model by converting high-precision numbers (32-bit) to lower-precision formats (4-bit, 8-bit). Think of it like compressing an image—the file gets smaller but you lose some quality. A 7.6GB model can shrink to 2.5GB with minimal accuracy loss.

GGUF (GPT-Generated Unified Format)

A file format specifically designed for running language models efficiently on CPUs. It's optimized for consumer hardware and supports various quantization levels in a single standardized format.

Perplexity

A score measuring how "confused" the model is when predicting the next word. Lower is better. Think of it as the number of words the model is uncertain between—a perplexity of 2 means it's choosing between ~2 options. In quantization, it shows quality loss: +0.05 perplexity = barely noticeable quality drop.

Context Length/Context Window

The maximum number of tokens (word pieces) the model can "remember" at once. Phi-4 Mini supports 128K tokens (~96,000 words or ~200 pages). Longer contexts require more RAM for the KV cache.

Tokens

Text chunks the model processes—usually parts of words. "Hello world" $\approx$ 2-3 tokens. Models measure speed in tokens/second (t/s). Typical reading speed is $\sim$ 4-5 tokens/second, so 10 t/s feels instant.

Quantization Levels

Q4_K_M (4-bit quantization)

Compresses each parameter to 4 bits ($\sim$ 2.5GB file size). Excellent balance: 67% smaller than original with $\sim$ 99.5% quality retention. Best for most users with limited RAM.

Q5_K_M (5-bit quantization)

Slightly higher precision at 5 bits per parameter ($\sim$ 3.0GB file). Only 0.5GB larger than Q4 but maintains 99.9% quality. Recommended for production use.

Q8_0 (8-bit quantization)

High-quality 8-bit format ($\sim$ 4.1GB file). Near-perfect 99.99% accuracy retention. Required for 64K+ context lengths to maintain quality.

FP16 / F16 (16-bit floating point)

Half-precision format, the original model quality ($\sim$ 7.6GB). No compression loss but requires double the RAM of Q8.

Performance Metrics

RAM (Random Access Memory)

System memory required to load and run the model. Phi-4 needs base model size + KV cache. Example: Q5_K_M = 3GB model + 2GB cache at 16K context = 5GB total RAM.

KV Cache (Key-Value Cache)

Memory used to store conversation history so the model doesn't recompute previous tokens. Scales with context length: $\sim$ 0.25GB per 4K tokens for Q5, $\sim$ 0.35GB per 4K for Q8.

Inference Speed (Tokens/Second)

How fast the model generates text, measured in tokens per second (t/s). Q4: 10-15 t/s, Q5: 8-12 t/s, Q8: 5-8 t/s on 6-core CPU. Faster than human reading (4-5 t/s) feels responsive.

AVX2 (Advanced Vector Extensions 2)

CPU instruction set for parallel processing that speeds up AI calculations by 4-8x. All modern CPUs since 2015 support it. Essential for efficient GGUF model inference.

Technical Terms

Parameters

The learned weights in the neural network—Phi-4 Mini has 3.8 billion parameters. More parameters generally mean better quality but require more resources.

Model Weights

The numerical values (parameters) that determine how the model processes information. Quantization compresses these weights from 32-bit floats to 4-8 bit integers.

Llama.cpp

The underlying inference engine that runs GGUF models efficiently on CPUs. Powers tools like Ollama and supports AVX2 optimizations.

Ollama

A user-friendly tool for running GGUF models locally, built on llama.cpp. Handles model loading, API serving, and optimization automatically.

Quick Reference

Term	Simple Meaning	Why It Matters
GGUF	Optimized model file format	Runs efficiently on your CPU
Q4/Q5/Q8	Compression levels	Trade-off between file size and quality
Context	Model's "memory" span	Determines how much text it processes at once
RAM	System memory needed	Limits what models you can run
t/s	Text generation speed	Higher = faster responses
Perplexity	Quality measurement	Lower = better accuracy